

C# BACKEND TRACK | OOP PHASE

Task 1

System Analysis Task

Think about the business. Extract the models. Assign the sources.

Codeline Bootcamp · Backend Phase · OOP Session 3 Task

SOURCE TAG LEGEND

Every property of every class model gets its value from somewhere. Before writing any code, you must decide **where** each value comes from. Use one of these five source tags to label each property in your analysis:

System Generated	System Generated	The system assigns this value automatically. The user never types it. Example: an auto-incremented ID calculated as List.Count + 1.
User Input	User Input	The user types this value directly at the keyboard when prompted by the system.
Default Value	Default Value	A fixed value set by the code when the object is created. The user never chooses it. Example: isBooked = false, status = "Active".
Calculated	Calculated	Derived automatically from another object already in memory — no user prompt. Example: appointment date copied from a chosen slot, fee copied from a doctor.
From List	From List	The user selects a value from a list of options displayed on screen. Example: choosing a doctor ID from a printed list of doctors.

Domain 1 — Library Management System

Extract everything yourself

Read the following description of a real library carefully. Your job is to go through the 6-phase process and fill in all the blank tables below. Nothing has been done for you — you identify the entities, you decide the properties, and you assign the source tag for each one.

A public library allows community members to register and borrow books. Each book has a title, author, genre, and a fixed number of physical copies in stock. A registered member can borrow an available book copy and must return it within 14 days. If the book is returned late, the system calculates a fine based on how many days overdue it is. A member can also reserve a book that is currently fully borrowed so they are notified when it becomes available. The librarian needs to see all currently active loans, view all overdue loans sorted by how late they are, and get a report of the most-borrowed books ranked by total borrow count.

Step 1 — Identify the Entities (Phases 1 & 2)

From the description above, identify the main real-world objects (classes) the system must track. Write each entity name, its properties, the data type for each, and the source tag that describes how each property gets its value.

Entity Name: <u>Member</u>		
Property Name	Data Type	Source Tag
<u>memberId</u>	<u>int</u>	<u>System Generated</u>
<u>memberName</u>	<u>string</u>	<u>User Input</u>
<u>memberPhone</u>	<u>string</u>	<u>User Input</u>
<u>memberEmail</u>	<u>string</u>	<u>User Input</u>
<u>registrationDate</u>	<u>string</u>	<u>System Generated</u>
<u> </u>	<u> </u>	<u> </u>
<u> </u>	<u> </u>	<u> </u>

Entity Name: <u>Book</u>		
Property Name	Data Type	Source Tag
<u>bookId</u>	<u>int</u>	<u>System Generated</u>
<u>title</u>	<u>string</u>	<u>User Input</u>
<u>author</u>	<u>string</u>	<u>User Input</u>
<u>genre</u>	<u>string</u>	<u>User Input</u>
<u>totalCopies</u>	<u>int</u>	<u>User Input</u>
<u>availableCopies</u>	<u>int</u>	<u>Calculated</u>
<u>totalBorrowCount</u>	<u>int</u>	<u>Default Value</u>

Entity Name: <u>Loan</u>		
Property Name	Data Type	Source Tag
<u>loanId</u>	<u>int</u>	<u>System Generated</u>
<u>memberId</u>	<u>int</u>	<u>From List</u>
<u>bookId</u>	<u>int</u>	<u>From List</u>
<u>borrowDate</u>	<u>string</u>	<u>System Generated</u>
<u>dueDate</u>	<u>string</u>	<u>Calculated</u>
<u>returnDate</u>	<u>string</u>	<u>Default Value</u>
<u>status</u>	<u>string</u>	<u>Default Value</u>

Entity Name: <u>Reservation</u>		
Property Name	Data Type	Source Tag
<u>reservationId</u>	<u>int</u>	<u>System Generated</u>
<u>memberId</u>	<u>int</u>	<u>From List</u>
<u>bookId</u>	<u>int</u>	<u>From List</u>
<u>reservationDate</u>	<u>string</u>	<u>System Generated</u>
<u>isNotified</u>	<u>bool</u>	<u>Default Value</u>
<u></u>	<u></u>	<u></u>
<u></u>	<u></u>	<u></u>

If you identify more than 4 entities, continue on the back of this sheet or add extra rows above.

Step 2 — Describe the Entity Relationships (Phase 1 continued)

How do the entities connect to each other? Describe the full cycle in plain English — who does what, which entity gets created, and what changes in other entities as a result.

Write the relationship cycle here:

A member registers in the library system. A member can borrow an available book, so the system creates a Loan record and updates the available copies. The system calculates the due date. If the book is returned late, the fine is calculated. If no copies are available, the member can reserve the book and will be notified later.

Step 3 — List the Services (Phase 4)

From the description, identify every service (function) the system needs. Write at least 8 services — give each a name and a one-line description of what it does.

#	Service Name	What it does
1	RegisterMember	Create a new member in the library system.
2	AddBook	Add a new book with its details and copies.
3	ViewBooks	Display all books in the library.
4	BorrowBook	Allow a member to borrow an available book.
5	ReturnBook	Return a borrowed book and update availability.
6	ReserveBook	Reserve a book when all copies are borrowed.
7	CalculateFine	Calculate fine for overdue books.
8	ViewActiveLoans	Display all active loans.
9	ViewOverdueLoans	Display overdue loans sorted by late days.
10	ViewBorrowReport	Display the most borrowed books.

Domain 2 — Hotel Management System

Assign the source tags

The class models and all their properties are already defined for you below. Your job is to **fill in the Source Tag column** for every single property. Use the five source tags from the legend on page 1: System Generated, User Input, Default Value, Calculated, From List.

A hotel allows guests to register and book rooms for a stay. The hotel has different room types (Single, Double, Suite) each with its own nightly price. Each room can only have one active booking at a time. When a guest checks in, a check-in record is created. The total price of a stay is calculated from the number of nights and the room's nightly rate. When a guest checks out, the room becomes available again. A booking can also be cancelled before check-in, which frees the room immediately.

Your task: For each property in the tables below, write the correct source tag in the empty column on the right. Think carefully — ask yourself: does the user type this? does the system generate it? is it copied from another object? does it have a fixed starting value? does the user pick it from a list?

Guest		
Property Name	Data Type	Source Tag (fill in)
<code>guestId</code>	int	_____ System Generated
<code>guestName</code>	string	_____ User Input
<code>guestPhone</code>	string	_____ User Input
<code>guestEmail</code>	string	_____ User Input
<code>guestNationality</code>	string	_____ User Input
<code>registrationDate</code>	string	_____ System Generated

Room		
Property Name	Data Type	Source Tag (fill in)
roomId	int	_____ System Generated _____
roomNumber	string	_____ User Input _____
roomType	string	_____ From List _____
floorNumber	int	_____ User Input _____
nightlyPrice	decimal	_____ User Input _____
isAvailable	bool	_____ Default Value _____

Booking		
Property Name	Data Type	Source Tag (fill in)
bookingId	int	_____ System Generated _____
guestId	int	_____ From List _____
roomId	int	_____ From List _____
checkInDate	string	_____ User Input _____
checkOutDate	string	_____ User Input _____
numberOfNights	int	_____ Calculated _____
totalPrice	decimal	_____ Calculated _____
status	string	_____ Default Value _____

CheckInRecord		
Property Name	Data Type	Source Tag (fill in)
recordId	int	System Generated
bookingId	int	From List
guestId	int	Calculated
roomId	int	Calculated
actualCheckIn	string	System Generated
actualCheckOut	string	Default Value
finalAmount	decimal	Calculated

Step 2 — Describe the Entity Relationships

Now that you have studied the four class models, describe in your own words how they connect to each other. Who creates what? What changes when a booking is made? What happens when a guest checks out?

Write the relationship cycle here:

A guest registers in the hotel system and books an available room. A Booking record is created and the total price is calculated. The room becomes unavailable during the booking period. When the guest checks in, a CheckInRecord is created. When the guest checks out, the final amount is saved and the room becomes available again.

Step 3 — List the Services

Based on the description and the class models above, list all the services (functions) this hotel system needs. Give each a name and a one-line description.

#	Service Name	What it does
1	AddNewRoom	Create a new room and add it to the hotel system.
2	UpdateRoom	Update room information such as type or price.
3	ViewAllRooms	Display all rooms and their availability.
4	RegisterGuest	Create a new guest record.
5	BookRoom	Create a booking for selected guest and room.
6	CancelBooking	Cancel booking and make the room available.
7	CheckIn	Create a check-in record when guest arrives.

#	Service Name	What it does
8	CheckOut	Complete guest checkout and update room availability.
9	ViewBookings	Display all booking records and their current status.